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# Optimizing university course timetabling for metaverse integration: a human-centered decision model in medical education

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## Abstract

**Background** University Course Timetabling Problems (UCTTP) in medical schools have become more complex with the integration of metaverse technologies such as virtual and augmented reality. These immersive environments enhance medical education but also introduce new scheduling constraints involving specialized hardware, virtual classrooms, and simulation sessions. Addressing this need, the present study develops a human-centered (faculty-focused) optimization framework that aligns scheduling decisions with instructors' behavioral intentions to adopt metaverse-based teaching methods.

**Methods** A binary integer linear programming model was formulated to jointly assign regular and metaverse courses while satisfying institutional constraints. Professor-specific behavioral intention weights were derived using the Analytic Hierarchy Process (AHP) informed by constructs identified through Structural Equation Modeling (SEM). Two heuristic approaches—the Greedy Reassignment and Assignment for Professor Equity (GRAPE) and Simulated Annealing (SA)—were developed to solve the model. The parameters of SA were optimized using the Taguchi Design of Experiments method. Computational experiments were conducted on 45 synthetically generated instances of varying sizes.

**Results** The results show that the GRAPE algorithm provides rapid feasible solutions, whereas the SA algorithm yields higher solution quality, particularly for large and complex problem instances. The Taguchi analysis indicated that the cooling rate and number of iterations significantly influence performance, achieving an excellent model fit ($R^2 = 0.998$). Overall, the SA method consistently produced near-optimal schedules with low relative error values while maintaining reasonable computation times.

**Conclusions** This study introduces the first optimization framework that integrates behavioral intention modeling into course timetabling for metaverse-based education. The proposed approach enables institutions to allocate metaverse courses to instructors who demonstrate higher readiness to adopt immersive teaching technologies. These findings support the design of adaptive, human-centered scheduling systems aligned with faculty readiness. The current framework focuses on faculty-level behavioral factors; student-related dimensions, such as technological access and cohort preparedness, are acknowledged as important areas for future research.

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**Keywords** University course timetabling, Educational technology integration, Metaverse in medical education, Integer linear programming, Analytic hierarchy process, Simulated annealing

## Introduction

### Background

The University Course Timetabling Problem (UCTTP) addresses the challenge of assigning students, instructors, and courses to specific timeslots and classrooms while ensuring that the resulting schedules for students and faculty comply with technical constraints [1]. The timetabling problem fundamentally represents a resource allocation challenge, and it is widely recognized that such allocation tasks are combinatorial in nature and classified as NP-complete problems [2]. The majority of NP-complete problems are difficult to solve optimally and efficiently due to the vastness of their search spaces, which expand exponentially with the increase in the number of variables [3]. Due to this complexity, exact solution approaches become computationally infeasible and require the adoption of heuristic and metaheuristic optimization strategies.

At the same time, medical education is experiencing a digital transformation with the emergence of the metaverse, which integrates virtual reality (VR), augmented reality (AR), and other immersive technologies within shared virtual environments [4, 5]. A recent large-scale bibliometric analysis conducted by Damar and Koksalmis draws attention to metaverse applications in healthcare and highlights the growing research interest in this area within medical education [6]. Within the metaverse context, medical students are able to engage deeply with highly realistic and interactive virtual environments that replicate clinical settings, including virtual hospitals, operating rooms, and patient interactions [7]. These interactive and immersive environments provide students with a flexible and safe learning space while supporting active participation, student-centered approaches, and personalized learning experiences. In this respect, they align with the constructivist pedagogical principles proposed by Lam et al. [8]. Empirical studies have also highlighted the pedagogical value of these immersive environments. For example, Mergen et al. demonstrated that immersive VR-based simulations significantly improved medical students' perceived self-efficacy in performing skin cancer screenings [9]. Furthermore, Hammouda et al. demonstrated that VR anatomy simulations significantly enhance medical students' understanding of anatomical structures, alleviate their anxiety, and are strongly supported by students as a complementary tool in anatomy education [10]. In parallel, Campos et al. demonstrated the positive effects of simulation-based education (SE) on learning. Their study emphasized that SE, delivered through tools such as virtual laboratories and online platforms, enhanced students' critical thinking, decision-making, and collaboration skills [11].

AR and VR technologies also contribute significantly to specific domains of medical education. In surgical training, doctors can use AR to rotate specific anatomical structures during brain and cardiac surgeries, allowing for better visualization in the process of performing, planning and explaining the surgery. In anatomy education, VR technology provides an immersive three-dimensional digital platform where students can examine and interact with realistic human body models [12]. Within classroom settings, AR tools, tools such as augmented reality pens can allow students to examine and learn concepts through three-dimensional images [13]. In medical imaging, AR enables three-dimensional scan data to be displayed directly on the patient's body to support visualization and surgical planning. Additionally, AR and VR technologies are used in team-based training by simulating situations that require coordination among healthcare professionals, supporting the development of communication and collaboration skills in settings such as emergency care [4]. Furthermore, simulation-based medical education presents opportunities to minimize risks for both patients and students, strengthen student competence and confidence, improve patient safety, and contribute to the reduction of healthcare costs over time [14].

This study aims to develop a mathematical model that assigns regular courses and metaverse-based courses together. Here, regular courses refer to courses conducted through traditional, face-to-face instruction in standard physical classrooms. In contrast, metaverse courses involve immersive technologies such as VR, AR, or simulation-based platforms and require specialized virtual or metaverse-enabled environments. The empirical framework of this study is grounded in the context of Turkish medical schools, reflecting their institutional structures, administrative practices, and emerging adoption of immersive technologies. An important feature of the study is the integration of weights into the mathematical model, denoted as $w(i,p)$, derived from an Analytic Hierarchy Process (AHP) based on a previous structural equation model (SEM) [15]. The SEM was designed to assess medical educators' intention to adopt metaverse technologies in teaching and its important constructs were used as criteria to calculate these weights. By placing these weights in the objective function of a binary integer programming model, the study aims to prioritize the assignment of metaverse courses to professors who show greater willingness and adaptability to immersive

teaching environments. This approach addresses emerging institutional scheduling needs as medical education increasingly incorporates metaverse-based learning methods.

The unique contribution of this study lies in integrating behavioral intention modeling with course scheduling optimization. By combining weights derived from an SEM and AHP, the proposed model extends traditional course scheduling approaches to account for faculty's behavioral intention to teach metaverse-based courses. This link between metaverse technology adoption and scheduling decisions enables more informed, personalized, and future-ready academic planning. To the best our knowledge, this is the first course planning model to combine regular and metaverse courses in a unified optimization framework driven by faculty weighting. As immersive technologies become increasingly central to medical education, this approach offers a novel mechanism for aligning instructional delivery with faculty readiness and institutional needs.

## Literature review

A bibliometric analysis was conducted to comprehensively examine studies addressing university course timetabling problems. Relevant publications were identified through a Scopus search using keywords related to timetabling, scheduling, and metaheuristics in higher education contexts. The search yielded 236 records. A co-occurrence analysis of author keywords using VOSviewer revealed that Simulated Annealing is among the most frequently applied metaheuristic techniques in university course timetabling research (Fig. 1), underscoring its central role in solving highly constrained scheduling problems. These findings also provide context for the methodological choices in this study, as the prominence of Simulated Annealing in the co-occurrence network highlights its central role in UCTTP research and supports its use as a primary optimization approach in the proposed framework.

This bibliometric mapping highlights the algorithmic orientation of existing UCTTP research and provides the foundation for examining how optimization-based scheduling studies have evolved. However, it does not capture the parallel stream of research focusing on behavioral readiness for emerging instructional technologies, which is examined separately below.

### *Classical metaheuristic approaches*

Metaheuristic algorithms such as Simulated Annealing (SA), Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Tabu Search (TS), and Harmony Search (HS) have been widely applied to the University Course Timetabling Problem (UCTTP). Bellio et al. [16] proposed a feature-guided SA for curriculum-based timetabling, while Sylejmani et al. [17] introduced a two-stage SA prioritizing feasibility before optimizing

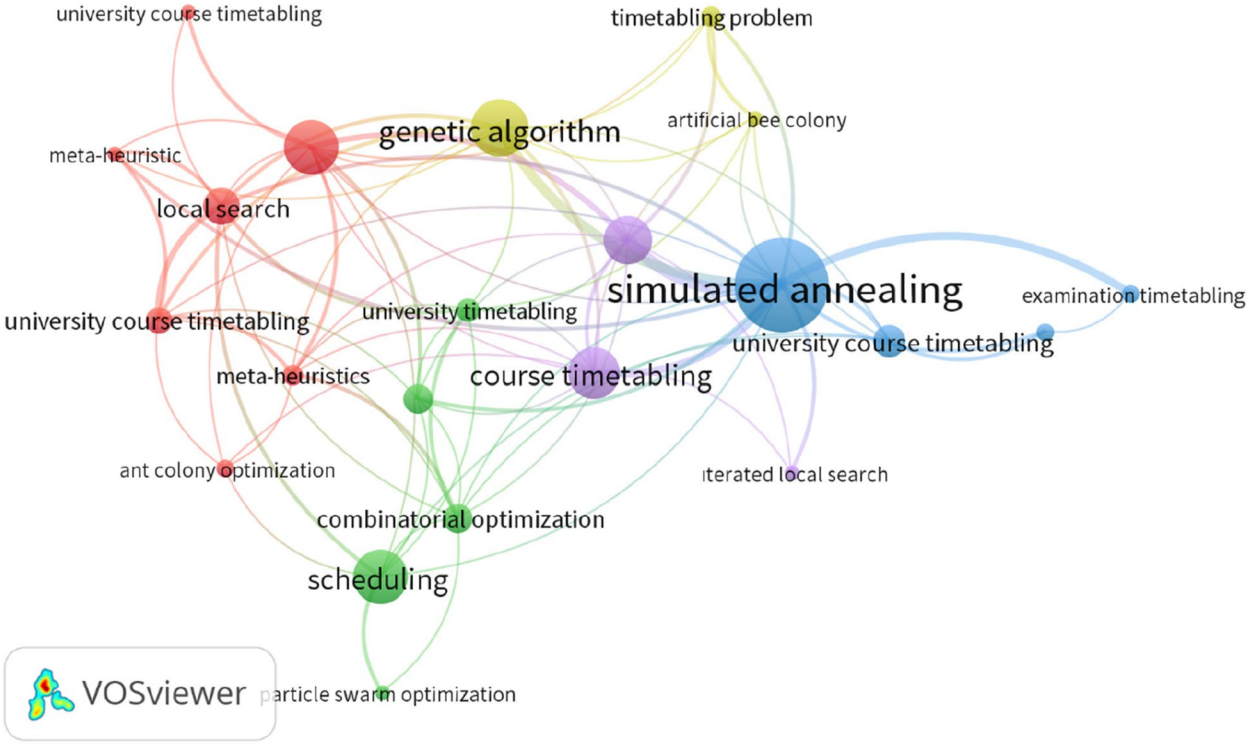



**Fig. 1** Co-occurrence analysis of author keywords

soft constraints. Hossain et al. [18] enhanced PSO using selective search mechanisms, and Mahlous and Mahlous [19] incorporated student preferences into a GA framework. HS and its modified variants have shown competitive scheduling performance [20]. Similarly, TS approaches have also demonstrated effective escape from local optima [21].

Although these methods improve solution quality and computational efficiency, instructors are generally modeled as interchangeable resources constrained only by eligibility and availability. Empirically grounded behavioral or technological readiness factors are not incorporated into the optimization structure.

#### *Hybrid and adaptive metaheuristic techniques*

To enhance robustness and scalability, hybrid and adaptive metaheuristic frameworks have been developed. Vianna et al. [22] combined Variable Neighborhood Search with TS, while Wong et al. [23] embedded TS within GA initialization. Other hybrid models include GA with Iterative Local Search [24] and Artificial Bee Colony with Hill Climbing [25]. Adaptive approaches improve search efficiency by dynamically modifying algorithm behavior, such as intelligent mutation in GA [26] and adaptive neighborhood selection in SA [27].

Despite their algorithmic sophistication, these studies remain computationally oriented. Institutional context, pedagogical considerations, and faculty-level technology adoption readiness are not explicitly embedded in scheduling formulations.

#### *Metaverse adoption in medical education*

A growing body of research has examined metaverse adoption in healthcare and medical education using established behavioral intention frameworks such as TAM, UTAUT2, and SEM. Studies have identified determinants of technology adoption including perceived usefulness, perceived ease of use, personal innovativeness, perceived enjoyment, social influence, and facilitating conditions [28–30].

More advanced empirical approaches combining SEM with machine learning techniques further confirm the significant role of psychological and contextual constructs in shaping metaverse adoption [31, 32]. However, these studies primarily focus on explaining or predicting behavioral intention, and do not translate empirically derived readiness factors into institutional planning mechanisms such as course allocation or timetabling decisions.

#### *Research gap and positioning*

The literature thus reveals two largely independent research streams:

1. Optimization-based university course timetabling studies emphasizing computational performance and algorithmic efficiency.
2. Behavioral intention studies examining metaverse adoption in educational contexts.

While timetabling research advances algorithmic sophistication, instructors are typically modeled as homogeneous resources constrained only by eligibility and availability. Conversely, metaverse adoption studies empirically quantify behavioral readiness constructs but do not operationalize these constructs within institutional planning or scheduling mechanisms.

To clarify this methodological separation, Table 1 summarizes the positioning of related work.

As shown in Table 1, prior research has addressed either behavioral modeling or timetabling optimization, but not their structured integration. The present study bridges this gap by embedding SEM-derived behavioral constructs into an AHP-based weighting mechanism and incorporating these weights into a binary integer programming timetabling model. To the best of our knowledge, such integration has not been previously reported in the context of metaverse-based medical education.

**Table 1** Conceptual positioning of the proposed framework within existing research streams

| Research Stream/ Study Type | Empirical behavioral intention modeling (SEM/TAM/UTAUT) | Course timetabling optimization | Integrated framework |
|---|---|---|---|
| Metaverse adoption studies in medical education | ✓ | ✗ | ✗ |
| Classical & hybrid UCTTP optimization studies | ✗ | ✓ | ✗ |
| This study | ✓ | ✓ | ✓ |

### Problem definition

The UCTTP needs advanced mathematical models to meet the diverse and complex needs of academic institutions. Instead of focusing only on a specific case, this paper proposes a general model that includes different scenarios. The proposed model is flexible and inclusive and can be adapted to various educational needs. In particular, the inclusion of metaverse-based courses in this framework makes it possible to adapt to modern teaching methods in medical education and to adapt faster to technological developments.

The problem aims to create an efficient timetable within the university by assigning courses to available classes,

time slots and days according to various constraints. This optimization process involves the development of a feasible schedule for regular courses, professors, and metaverse courses based on certain constraints. Accordingly, the model is described below:

- The term timeslot signifies a duration of one hour, encompassing nine such intervals within a day from 8:00 AM to 5:45 PM, occurring Monday through Friday.
- The courses are divided into regular courses and metaverse courses.
- Professors demonstrate a fit based on their field competencies by undertaking courses that overlap with their areas of expertise. In the case of metaverse-based courses, their technological aptitude and potential to teach such courses were taken into account. In this context, professors' level of intention to use metaverse technology was determined through a previous SEM, and professors with a high level of intention were prioritized to be assigned metaverse courses. Thus, professors with a high capacity to adapt to technological innovations were encouraged to take on this new generation of courses.
- The challenges of implementing medical education in the metaverse are multifaceted. These include limitations on internet bandwidth and device (e.g. smartphones or computers) access, restrictions on the availability of AR/VR hardware, health-related usage considerations, restrictions on platform selection and integration, guidelines on the use of anonymity, and regulations to protect intellectual property [5]. Additional concerns include cognitive overload, challenges related to the educational process, isolation and lack of social interaction, as well as motion sickness [33]. Addressing both hardware-related and cognitive challenges within the model is essential for ensuring that the developed framework accurately reflects real-world conditions. Accordingly, these challenges are systematically incorporated into the proposed optimization model through a combination of structural constraints and behavioral weighting mechanisms. A detailed mapping between these challenges and their corresponding model components is provided in the Binary Integer Programming subsection.

## Methodology

To address the university course scheduling problem in the context of metaverse-based education, an appropriate methodological framework is required. For this purpose, a multi stage approach has been developed. Figure 2 presents an overview of the proposed framework. The first step involves constructing a Structural Equation Model to identify the key factors influencing professors' intentions to use metaverse technologies based on the study by Damar and Koksalmis [15]. These factors are then used as criteria in the Analytic Hierarchy Process to derive professor weights. The resulting weights are incorporated into the objective function of a Binary Integer Programming (BIP) model. To solve the model, a greedy algorithm referred to as Greedy Reassignment and Assignment for Professor Equity (GRAPE) and a Simulated Annealing algorithm are employed. In addition, the Taguchi Design of Experiments (DoE) is used to determine the optimal parameters for the SA algorithm. A feasibility check is performed after the initial GRAPE solution, followed by a performance evaluation step comparing GRAPE and SA solutions. Each step of the methodology is described in detail in the following subsections.

### Structural equation modeling approach

Structural Equation Modeling refers to a suite of statistical methods used to analyze complex relationships among multiple independent and dependent variables, which may be either continuous or categorical. These variables can represent latent constructs (factors) or directly observed measures. SEM is also commonly known by several other terms, including causal modeling, causal analysis, simultaneous equation modeling, covariance structure analysis, path analysis, and confirmatory factor analysis. Notably, path analysis and confirmatory factor analysis are considered specific subtypes within the broader SEM framework [34].

SEM offers several advantages over traditional statistical methods. It enables the simultaneous analysis of multiple dependent variables and accommodates measurement errors in both independent and dependent variables. SEM allows for the concurrent estimation of latent variable structures and the relationships among them, thereby integrating factor analysis and path analysis within a unified framework. Moreover, it supports flexible measurement models, including those with indicators linked to multiple factors, and facilitates the assessment of model fit, providing a more comprehensive understanding of the data's underlying structure [35].

Although SEM provides standardized path coefficients that indicate the strength and significance of relationships among latent constructs, these coefficients are not directly structured for use as decision weights within an optimization framework. In this study, SEM is primarily employed to validate the behavioral model and to identify the relative importance of constructs influencing professors' intention to teach metaverse-based courses. The empirically supported constructs are then operationalized within an AHP framework to derive normalized and consistent weights suitable for integration into the binary

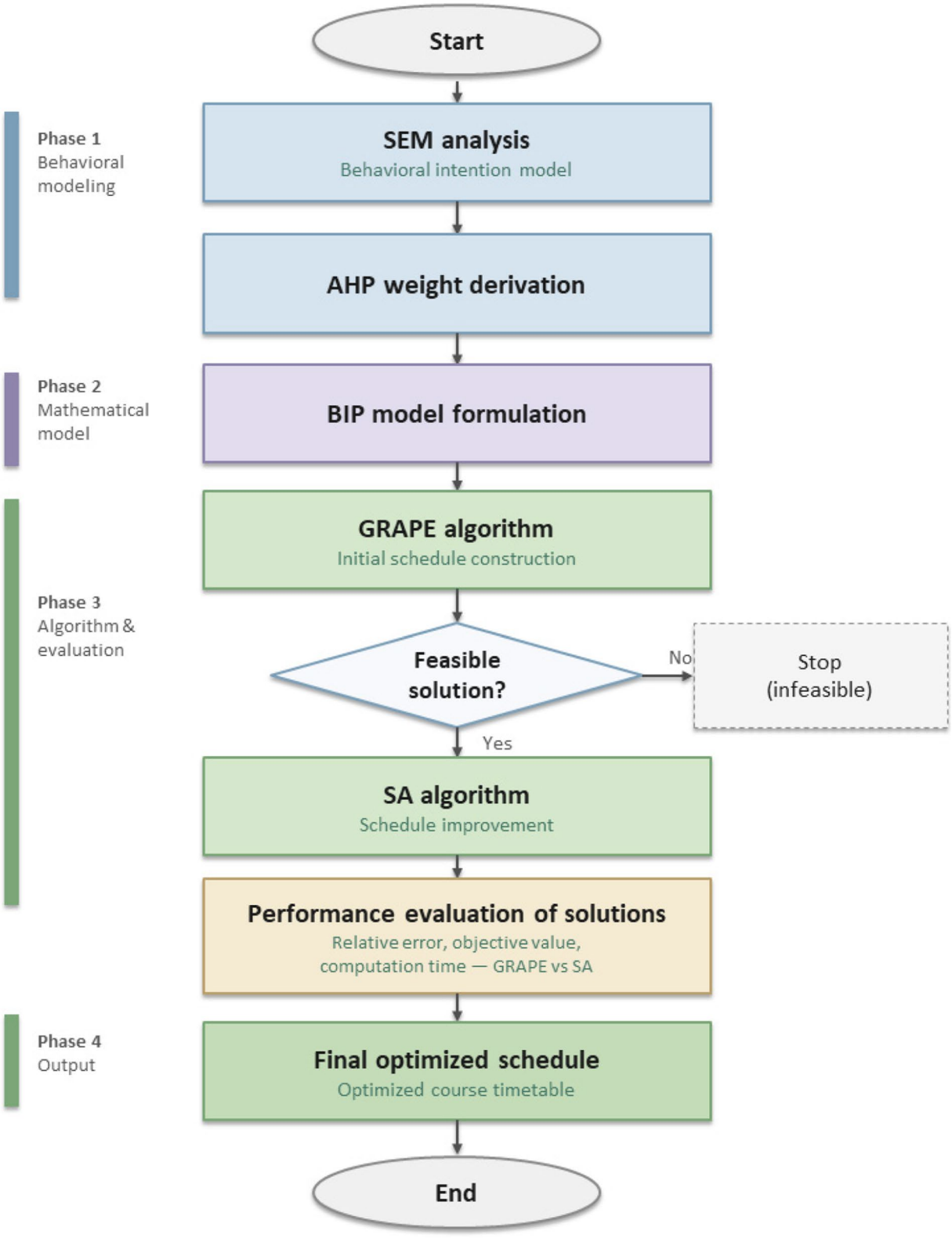



**Fig. 2** Flowchart of the proposed multi-stage methodology. The framework includes SEM-based behavioral modeling, AHP weight derivation, Binary Integer Programming (BIP) formulation, GRAPE-based initial solution construction, feasibility checking, SA-based improvement, and performance evaluation

integer programming model. In this way, SEM serves as a theoretical and empirical foundation, while AHP translates validated behavioral criteria into actionable decision parameters for the scheduling problem. The methodological validity of this integration relies on the interpretation of standardized SEM total effects as indicators of the relative contribution of each construct to behavioral intention. Since these coefficients are expressed on a common scale, normalizing them to sum to unity preserves their proportional differences and allows them to be used as weights consistent with the ratio-scale prioritization logic of AHP. In this way, the SEM–AHP integration provides an empirically grounded alternative to traditional pairwise comparisons while maintaining the structural rationale of the AHP framework.

#### Analytic hierarchy process approach

The Analytic Hierarchy Process is a decision-making methodology designed to address problems involving multiple objectives and criteria. It utilizes a systematic pairwise comparison technique to establish a prioritized ranking among a set of alternatives based on relative preferences [36]. It is particularly effective in structuring complex decisions by decomposing them into a hierarchy consisting of a goal, evaluation criteria, and alternatives [37]. The method utilizes ratio-scale judgments through pairwise comparisons, and these judgments are mathematically synthesized by solving an eigenvalue problem to derive priority weights [37].

In AHP, judgments are collected in the form of pairwise comparisons, which are used to construct a reciprocal matrix $A = [a_{ij}]$, where each entry $a_{ij}$ denotes the relative importance or preference of element *i* over element *j* [38]. By definition, the matrix must satisfy the condition given in Eq. (1).

$$a_{ji} = \frac{1}{a_{ij}}, a_{ii} = 1 \tag{1}$$

This ensures the matrix is positive and reciprocal, a foundational property in AHP.

If the pairwise comparisons are perfectly consistent, then the following condition holds for all *i*, *j*, *k* as shown in Eq. (2).

$$a_{ij}.a_{jk} = a_{ik} \tag{2}$$

In such cases, there exists a priority vector $w = [w_1, w_2, \ldots, w_n]^T$, as expressed in Eq. (3).

$$a_{ij} = \frac{w_i}{w_j} \tag{3}$$

This leads to the matrix form presented in Eq. (4).

$$A = \begin{bmatrix} 1 & \frac{w_1}{w_2} & \cdots & \frac{w_1}{w_n} \\ \frac{w_2}{w_1} & 1 & \cdots & \frac{w_2}{w_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{w_n}{w_1} & \frac{w_n}{w_2} & \cdots & 1 \end{bmatrix} \tag{4}$$

To estimate the ratio-scale weights from real-world (and potentially inconsistent) judgments, the principal right eigenvector of the matrix $A$ is obtained by solving the eigenvalue equation presented in Eq. (5).

$$Aw = \lambda_{max} w \tag{5}$$

The eigenvector *w* corresponding to the maximum eigenvalue $\lambda_{max}$ is then normalized, as shown in Eq. (6).

$$\sum_{i=1}^{n} w_i = 1 \tag{6}$$

Saaty [38] showed that if the matrix is fully consistent, then $\lambda_{max} = n$; otherwise, $\lambda_{max} > n$, and the Consistency Index (CI) can be used to quantify the degree of inconsistency, as shown in Eq. (7).

$$CI = \frac{\lambda_{max} - n}{n - 1} \tag{7}$$

This formulation ensures that even when small inconsistencies are present, reliable ratio-scale weights can still be extracted.

#### Illustrative study

An illustrative example is created to demonstrate the integration of SEM and AHP methodologies in the proposed optimization framework. This example aims to show how important latent constructs identified through SEM are transformed into decision criteria for AHP and then used to derive professor weights for metaverse courses, denoted as *w(i,p)*. In the example, professors eligible to teach metaverse courses are treated as alternatives and criteria are identified from SEM analysis based on constructs with statistically significant total effects. These constructs are then normalized to generate the relative weights used to make pairwise comparisons between eligible professors for each criterion.

Assume that professors a, b and c are eligible for metaverse courses. Here these professors will be used as alternatives. For these alternatives, pairwise comparison matrices will be created for each criterion. For example, the pairwise comparison matrix for the criterion "imagination" is given in Table 2. The pairwise comparison values represent the randomly assigned values between the three professors regarding their suitability for the "imagination" criterion. A scale from 1 to 9 was used to

**Table 2** Pairwise Comparison matrix with respect to the imagination

| | Professor a | Professor b | Professor c | Weights |
|---|---|---|---|---|
| Professor a | 1 | 3 | 5 | 0.633 |
| Professor b | 1/3 | 1 | 3 | 0.260 |
| Professor c | 1/5 | 1/3 | 1 | 0.106 |

**Table 3** Normalized weight of the criteria

| Criteria | Total effects | Normalized total effects |
|---|---|---|
| Imagination | 0.218 | 0.175 |
| Perceived trialability | 0.128 | 0.102 |
| Technology anxiety | 0.126 | 0.101 |
| Technology readiness | 0.073 | 0.058 |
| Perceived usefulness | 0.491 | 0.393 |
| Perceived ease of use | 0.213 | 0.170 |

represent the importance of one professor over the other. On the scale, 1 means equal importance, 3 means moderate importance, 5 means strong importance, 7 means very strong importance and 9 means extreme importance. The weight of each criterion is shown in Table 3.

So, the relative scores of professor a, professor b, and professor c for imagination can be calculated as, professor a, (0.633*0.175 = 0.111), professor b, (0.260*0.175 = 0.046), professor c, (0.106*0.175 = 0.019). Similarly, the relative scores for the remaining criteria are calculated in the same manner. Accordingly, the professor weight (PW) for each professor p is computed using Eq. (8).

$$PW_p = \sum_{j=1}^{6} C_j b_{pj} \tag{8}$$

where $C_j$ is the relative weightage for the criterion j, $b_{pj}$ is the relative weightage for professor p with respect to j$^{th}$ criterion, $PW_p$ professor weight for professor p.

For example, in the aforementioned example, $C_1$ represents the weight assigned to the imagination criterion and is equal to 0.175. $b_{a1}$ denotes the relative weight for professor a with respect to first criterion, which is 0.633. $PW_a$ is then computed using Eq. (9).

$$\begin{aligned} PW_a =& 0.175 * 0.633 + 0.102 * b_{a2} \\ &+ 0.101 * b_{a3} + 0.058 * b_{a4} \\ &+ 0.393 * b_{a5} + 0.17 * b_{a6} \end{aligned} \tag{9}$$

In this example, six constructs were identified to measure intention to use and these constructs were evaluated as criteria. Using the AHP approach, 15 pairwise comparison questions are needed to calculate the relative weights of these criteria. However, when the SEM approach is used, there is no need to create such pairwise comparison matrices since the participants' responses are obtained directly through Likert scale items. Therefore, the SEM approach was preferred in this study in order to obtain the criteria weights in a more objective and practical way. Also, hypothetical values are used in this illustrative example to better demonstrate the SEM-AHP process. However, it is important to consider how such comparisons would be evaluated in real-world scenarios. In practical applications, pairwise comparisons between professors based on specific criteria could be conducted by academic decision-makers, such as department chairs, course scheduling coordinators, or curriculum committees.

## Mathematical model

### *Sets*

*C = {1,2,…,n}:* Set of regular courses.

*M = {1,2,…,m}:* Set of metaverse courses.

*R:* Set of rooms for regular courses.

*Rm:* Set of rooms for metaverse courses.

*T:* Set of available timeslots for both regular courses and metaverse courses.

*D = {1,2,…,l}:* Set of days.

*P = {1,2,…,k}:* Set of professors.

### *Decision variables*

- $\boldsymbol{x_{irtd}}$ is a binary variable representing whether course i is scheduled in room r at timeslot t on day d.
- $\boldsymbol{y_{ip}}$ is a binary variable representing whether course *i* is taught by professor *p*.
- $\boldsymbol{w_{iptd}}$ is a binary variable representing professor *p* teaches course *i* at timeslot *t* on day *d*.

### *Parameters*

*S(i):* Shows the number of students/students enrolled in each course.

*B(i, r):* Binary parameter indicating eligibility of rooms for metaverse courses.

*C(r):* Shows the capacity of each room.

*A(r, t, d):* Binary parameter indicating room availability for metaverse courses.

*Z(i, p):* Binary parameter indicating professor–course eligibility according to disciplinary expertise and institutional qualification requirements.

*w(i, p):* The weight reflects the intention to use metaverse technology. These weights were derived based on the previously conducted SEM and were calculated based on data to measure the intention of medical physicians to use metaverse-based applications.

*LB(p):* Minimum number of courses required to be assigned to professor p. This parameter is institution-specific; in the context of Turkish medical schools it is set to 1, reflecting the standard full-time faculty workload policy. In institutional settings that permit zero teaching

load, LB(p) can be set to 0, or any appropriate non-negative integer for heavier minimum loads.

***A binary integer programming formulation***

$$MaxZ = \sum_{i\in M}\sum_{p\in P} y_{ip} * w\,(i,p) \tag{10}$$

$$\sum_{r\in R\cup Rm}\sum_{t\in T}\sum_{d\in D} x_{irtd} = 1 \forall i \in C\cup M \tag{11}$$

$$\sum_{i\in C\cup M} x_{irtd} \le 1 \forall r \in R\cup Rm, \quad \forall t\in T, \forall d\in D \tag{12}$$

$$\sum_{i\in C\cup M} y_{ip} \ge LB(p) \forall p \in P \tag{13}$$

$$\sum_{p\in P} y_{ip} = 1 \forall i \in C\cup M \tag{14}$$

$$\sum_{i\in C\cup M} w_{iptd} \le 1 \forall p \in P, \quad \forall t\in T, \forall d\in D \tag{15}$$

$$w_{iptd} \le \sum_{r\in R\cup R_m} x_{irtd} \forall i \in C\cup M, \forall p\in P, \quad \forall t\in T, \forall d\in D \tag{16}$$

$$w_{iptd} \le y_{ip} \forall i \in C\cup M, \quad \forall p\in P, \forall t\in T, \forall d\in D \tag{17}$$

$$w_{iptd} \ge \sum_{r\in R\cup R_m} x_{irtd} + y_{ip} - 1 \forall i \in C\cup M, \quad \forall p\in P, \forall t\in T, \forall d\in D \tag{18}$$

$$\sum_{t\in T}\sum_{d\in D} x_{irtd} = B\,(i,r)\,\forall i\in M, \forall r\in Rm \tag{19}$$

$$\sum_{d\in D} x_{irtd} * S\,(i) \le C\,(r) * B\,(i,r) \quad \forall i\in M, \forall r\in Rm, \forall t\in T \tag{20}$$

$$\sum_{d\in D} x_{irtd} * S\,(i) \le C\,(r)\,\forall i\in C, \forall r\in R, \forall t\in T \tag{21}$$

$$\sum_{t\in T}\sum_{d\in D} x_{irtd} = 0 \forall i\in C, \forall r\in Rm \tag{22}$$

$$\sum_{t\in T}\sum_{d\in D} x_{irtd} = 0 \forall i\in M, \forall r\in R \tag{23}$$

$$\sum_{i\in M} x_{irtd} \le A\,(r,t,d)\,\forall r\in Rm, \forall t\in T, \forall d\in D \tag{24}$$

$$y_{ip} \le Z\,(i,p)\,\forall i\in C\cup M, \forall p\in P \tag{25}$$

$$x_{irtd}, y_{ip}, w_{iptd} \text{binary} \forall i\in C\cup M, \forall r\in R\cup Rm, \forall t\in T, \forall p\in P \tag{26}$$

The objective function (10) aims to prioritize the assignment of metaverse courses to professors with a higher weight, in order to ensure that these courses are conducted in the most efficient way. Constraint (11) ensures that each course, whether regular or metaverse-based, is scheduled exactly once in a specific room, time slot, and day. Constraint (12) prohibits a course from being assigned to multiple rooms simultaneously during the same time slot and day. Constraint (13) guarantees that each professor is assigned to teach at least LB(p) courses in the overall schedule. This reflects the common workload structure in Turkish medical schools, where full-time faculty members typically teach at least one course per semester, hence LB(p) = 1 for all p in this study. However, this constraint is context-dependent and can be relaxed or omitted in institutional settings that allow zero teaching load for faculty members by setting LB(p) = 0. The proposed model is flexible and can be easily adapted to such alternative scenarios. Constraint (14) enforces that each course is assigned to exactly one professor. Constraint (15), (16), (17), and (18) prevents any professor from being assigned to two overlapping courses within the same time and day. Constraint (19) ensures that metaverse courses are only scheduled in metaverse-eligible rooms. Constraint (20) restricts the total number of students in each metaverse-designated room from exceeding its capacity, considering the student size of each scheduled metaverse course. Constraint (21) applies the same room capacity restriction to regular courses. Constraint (22) prevents regular courses from being scheduled in metaverse-designated rooms. Conversely, constraint (23) prohibits metaverse courses from being assigned to rooms for regular courses. Constraint (24) enforces the use of only time slots and days during which metaverse-designated rooms are available, based on predefined availability. Constraint (25) ensures that professors are only assigned to courses for which they are eligible, according to subject expertise or institutional criteria. Finally, constraint (26) defines all decision variables as binary.

Although the challenges of metaverse-based medical education include factors such as hardware availability, platform access, and cognitive or usability-related

considerations, these aspects are operationalized in the proposed model through structural and resource-based constraints rather than as explicit psychological variables. In particular, limitations related to technological infrastructure and hardware availability are captured through constraints that restrict metaverse courses to metaverse-eligible rooms and available time slots (Constraints 19 and 24), while metaverse room capacity constraints limit the number of participants in immersive environments (Constraint 20). Eligibility constraints further ensure that metaverse courses are assigned only to professors who meet predefined technical or institutional requirements (Constraint 25). Cognitive factors associated with metaverse usage, such as technology anxiety or perceived difficulty, are not directly imposed as hard constraints; instead, their influence is incorporated via professor-specific weights derived from the Structural Equation Model, which are embedded in the objective function (Constraint 10). This abstraction allows the model to remain computationally tractable while still reflecting key practical limitations of metaverse-based instruction.

## Solution methods

### *Greedy reassignment and assignment for professor equity (GRAPE)*

The GRAPE algorithm is a constructive heuristic designed to rapidly generate a feasible initial timetable while respecting all hard constraints of the model. The algorithm operates in three main phases following initialization. First, metaverse courses are assigned using a priority-based strategy, where eligible professors are ranked according to their metaverse teaching weights and courses are scheduled only in metaverse-compatible rooms and time slots that satisfy availability and capacity constraints. Second, regular courses are assigned to professors who remain unassigned, ensuring minimum workload coverage when this policy is imposed, by selecting feasible room–time combinations that respect eligibility and availability constraints. Finally, a repair phase is applied to address any remaining unassigned professors. This phase first attempts equity-preserving reassignment from professors with multiple course allocations and, if necessary, applies a forced relocation step as a last resort. The repair process iterates until either all professors satisfy the minimum-load requirement or no further feasible reassignment or relocation is possible. By keeping room and time assignments fixed during reassignment, GRAPE preserves feasibility with respect to room compatibility, availability, and capacity constraints, while efficiently improving workload coverage. The pseudocode of the GRAPE algorithm is presented in Table 4.

Sets:

C, M: Regular Courses, Metaverse Courses
R, Rm: Regular Rooms, Metaverse Rooms
*T, D: Timeslots, Days*
*P: Professors*

*Parameters:*

S(c), Sm(m): Class sizes (Student count)
*wp(p):Professor weights*
*LB(p):Institution-specific minimum number of courses assigned to professor p*
*Ccapacity(r):Regular Room Capacity*
*Cm_capacity(rm): Metaverse Room Capacity*

*Matrices:*

*Z(i,p):Professor-Course Eligibility Matrix (1 if eligible, 0 else)*
*B(m,rm): Metaverse Room Eligibility Matrix*
*A(rm,t,d): Metaverse Room Availability Matrix*

From a computational perspective, GRAPE is a polynomial-time heuristic. In the worst case, its runtime grows proportionally with the number of courses, professors, rooms, and available time slots, as it performs greedy feasibility checks over these sets during assignment and reassignment. Consequently, its expected computational complexity is $O((|C|+|M|)\cdot|P|\cdot|R|\cdot|T|\cdot|D|)$, which is consistent with the very low execution times observed in the computational experiments.

### *Simulated annealing algorithm*

SA a popular technique for finding global or near-global optima of complex cost functions. It is widely recognized for its rapid convergence behavior and straightforward implementation [39]. Its robustness lies in its ability to escape local optima by probabilistically accepting worse solutions based on an acceptance probability $e^{\frac{-\Delta\theta}{T}}$, where $\Delta$ is the difference in objective function values between the current and candidate solutions, and $T$ is a temperature control parameter. The algorithm models the objective function as an energy metric, which is iteratively minimized through a controlled reduction in temperature, mirroring the behavior of physical annealing processes [40]. When T is high, the algorithm is more exploratory, accepting most moves (better or worse). As $T$ decreases, the algorithm becomes more exploitative, rejecting worse moves more frequently. To prevent premature convergence to a local minimum, the algorithm starts with a relatively high $T$.

In the context of this problem, the initial temperature is defined as a function of the objective function value of the initial schedule, making the starting temperature adaptive and reasonable. The SA algorithm then undergoes k temperature reductions according to the cooling

**Table 4** Pseudocode of the GRAPE algorithm

| | |
|---|---|
| **Phase 0 — Initialization** | |
| 1 | Sched ← ∅ |
| 2 | RoomAvail(x, t, d) ← 1 for all x ∈ (R ∪ Rm), t ∈ T, d ∈ D |
| 3 | P_avail(p, t, d) ← 1 for all p ∈ P, t ∈ T, d ∈ D |
| 4 | Load(p) ← 0 for all p ∈ P |
| 5 | Mark all courses as unassigned |
| **Phase 1 — Assign Metaverse Courses (Weighted)** | |
| 6 | SortedProf ← sort(P, wp, descending) |
| 7 | for each m ∈ M: |
| 8 | find any feasible (p, rm, t, d) such that: |
| 9 | p ∈ SortedProf, Z(m, p) = 1, P_avail(p, t, d) = 1, |
| 10 | rm ∈ Rm, B(m, rm) = 1, A(rm, t, d) = 1, RoomAvail(rm, t, d) = 1, |
| 11 | Cm_capacity(rm) ≥ Sm(m) |
| 12 | if found: |
| 13 | Sched[m] ← (rm, t, d, p) |
| 14 | P_avail(p, t, d) ← 0; RoomAvail(rm, t, d) ← 0; Load(p) + + |
| **Phase 2 — Assign Regular Courses for Minimum Load Coverage** | |
| 15 | SortedProf_Z ← sort(P, Σ Z(i, p) over i ∈ (C ∪ M), ascending) |
| 16 | for each p ∈ SortedProf_Z with Load(p) < LB(p): |
| 17 | find any feasible (c, r, t, d) such that: |
| 18 | c ∈ C is unassigned, Z(c, p) = 1, P_avail(p, t, d) = 1, |
| 19 | r ∈ R, RoomAvail(r, t, d) = 1, Ccapacity(r) ≥ S(c) |
| 20 | if found: |
| 21 | Sched[c] ← (r, t, d, p) |
| 22 | P_avail(p, t, d) ← 0; RoomAvail(r, t, d) ← 0; Load(p) + + |
| **Phase 3 — Repair (Reassignment + Forced Relocation)** | |
| 23 | Repeat until (all Load(p) ≥ LB(p)) OR (no successful move in a full pass): |
| 24 | successful_move ← False |
| 25 | for each orphan p with Load(p) < LB(p): |
| | *Step 3a — Reassignment from busy professors:* |
| 26 | find course i assigned to p' with Load(p') > LB(p') and fixed (room*, t*, d*) |
| 27 | such that Z(i, p) = 1 and P_avail(p, t*, d*) = 1 |
| 28 | if found: |
| 29 | reassign i from p' → p (keep room*, t*, d* unchanged) |
| 30 | update P_avail and Load; successful_move ← True; continue |
| | *Step 3b — Forced relocation (last resort):* |
| 31 | find course j assigned to any q with fixed (r, t, d) |
| 32 | such that Z(j, p) = 1 and P_avail(p, t, d) = 1 |
| 33 | if found: |
| 34 | relocate j from q → p (keep r, t, d unchanged) |
| 35 | update P_avail and Load; successful_move ← True |
| 36 | if successful_move = False: break |
| 37 | Return Sched |
| Notes | |
| 38 | Steps 3a–3b keep (room, t, d) fixed; thus room feasibility and capacity constraints remain satisfied |
| 39 | Step 3b may reduce the donor's load below its corresponding LB value; therefore, the repair process is repeated until no feasible move exists |

function $T_k = \alpha T_{k-1}$, where $\alpha$ is the cooling rate. At each temperature level, the algorithm explores the neighborhood of the current solution to identify better or potentially acceptable solutions.

The neighborhood search strategy in the SA algorithm is designed to prevent premature convergence to local optima by introducing controlled perturbations to the current solution. A swap-based neighborhood mechanism is employed, in which the professor assignments of two randomly selected courses are exchanged to generate a candidate neighbor. Since such swaps may violate scheduling constraints, each candidate neighbor is explicitly checked against all hard constraints of the model, including professor availability, room–time feasibility, room type compatibility, and capacity limits. If any constraint is violated, the candidate neighbor is discarded and a new neighbor is generated. For feasible neighbors, improvements in the objective function may arise, for example, when a metaverse course is reassigned to a professor with a higher weight. Non-improving but feasible neighbors may still be accepted according to the simulated annealing acceptance probability, allowing the algorithm to explore less favorable regions of the solution space and escape local optima.

The SA algorithm is employed as an improvement phase to refine the feasible initial timetable generated by the GRAPE heuristic. The algorithm begins with the GRAPE solution as the initial state and iteratively explores its neighborhood to reduce the objective function value while preserving feasibility. At each iteration, a neighboring solution is generated by performing a localized modification, such as swapping professor assignments between courses or reassigning a professor within the same room–time slot. To maintain feasibility, each candidate neighbor is validated against all hard constraints prior to evaluation. Feasible neighbors that improve the objective function are accepted, while non-improving neighbors may be accepted with a probability determined by the current temperature, allowing the algorithm to escape local optima. The temperature is gradually reduced according to a predefined cooling schedule, and the algorithm terminates when the stopping criterion is met. Through this feasibility-preserving neighborhood exploration and probabilistic acceptance mechanism, SA effectively improves solution quality while maintaining all scheduling constraints. The pseudocode of the SA algorithm is presented in Table 5.

k: Initial temperature coefficient

*α: Cooling rate*

*t_max: Maximum number of iterations*

*i_max: Maximum number of inner loop iterations*

**Table 5** Pseudocode of the Simulated Annealing algorithm

| | |
|---|---|
| **Initialization** | |
| 1 | Input: initial feasible schedule $S_{current}$ |
| 2 | Compute objective value: $\theta_{current} \leftarrow$ ObjectiveValue($S_{current}$) |
| 3 | $S_{best} \leftarrow S_{current}$; $\theta_{best} \leftarrow \theta_{current}$ |
| 4 | $T \leftarrow k \times \theta_{current}$ {initialise temperature} |
| **Main Loop** | |
| 5 | for $i = 1$ to $i_{max}$: |
| 6 | Repeat: |
| 7 | Generate $S_{new}$ by swapping two professor assignments in $S_{current}$ |
| 8 | Until $S_{new}$ satisfies all hard constraints |
| 9 | Compute: $\theta_{new} \leftarrow$ ObjectiveValue($S_{new}$) |
| 10 | $\Delta\theta \leftarrow \theta_{current} - \theta_{new}$ |
| 11 | if $\Delta\theta < 0$ or rand(0, 1) $< \exp^{(-\Delta\theta/T)}$: |
| 12 | Accept $S_{new}$ |
| 13 | if $S_{new}$ is accepted: |
| 14 | $S_{current} \leftarrow S_{new}$; $\theta_{current} \leftarrow \theta_{new}$ |
| 15 | if $\theta_{current} > \theta_{best}$: |
| 16 | $S_{best} \leftarrow S_{current}$ |
| 17 | $\theta_{best} \leftarrow \theta_{current}$ |
| 18 | $T \leftarrow T \times \alpha$ {cool temperature; $0 < \alpha < 1$} |
| **Output** | |
| 19 | Return $S_{best}$, $\theta_{best}$ |

## Results

The computational study conducted for the heuristic algorithms aims to evaluate the performance of the proposed methods across a wide range of problem sizes. The effectiveness and efficiency of the GRAPE and SA were tested using three different values for the number of regular courses ($n = 20, 41, 83$) and metaverse courses ($m = 7, 14, 28$), along with three different values for the number of professors. For each combination of $n$ and $m$, five instances were generated, resulting in a total of 45 problem instances. The binary integer programming model was implemented in Python using the PuLP optimization library and solved using the default CBC (Coin-or Branch and Cut) solver. All computational experiments were conducted in a cloud-based environment (Kaggle platform), which provides shared CPU resources and limited memory capacity. Default solver parameters were used unless otherwise specified, and the solver was allowed to run until optimality was reached or its internal termination criteria were triggered.

### Data generation

To evaluate the performance of the proposed course scheduling problem, synthetic datasets of different sizes are generated. The data generation process was designed taking into account the number of courses, rooms and other parameters specific to the problem. The datasets were categorized into three scales: small, medium and large.

The small dataset is the basis for obtaining larger datasets. The following steps were followed to create the dataset:

1. The total number of courses was randomly selected from a uniform distribution between 20 and 30.
2. A binomial distribution with a success probability of 0.2 was used to determine the number of metaverse courses. The remaining courses were designated as regular course sets (C).
3. The total number of rooms is based on the total number of courses, with a proportional relationship. The number of rooms was calculated as approximately one fifth of the total number of courses and rounded to the nearest whole number.
4. The number of rooms for metaverse courses was determined using a binomial distribution with a success probability of 0.2. At least one room was ensured for each metaverse course. The remaining rooms were designated as room sets for regular courses (R).
5. The number of timeslots was fixed at 9 and number of days was fixed at 5.
6. The enrollment numbers for each course (S(i)) are assumed to follow a normal distribution around a mean value. This reflects the tendency for most courses to have enrolments close to the mean; there are fewer courses with very high or very low enrolments.
7. Room capacities (C(r)) are also assumed to follow a normal distribution. This assumption implies that most rooms have capacities close to the average, with a smaller number of rooms having significantly higher or lower capacities.
8. Eligibility of rooms (B(i, r)), room availability (A(r, t, d)), and eligibility of professors (Z(i, p)) are modeled as binary parameters (0 or 1). In the synthetic data generation process, these parameters are randomly generated to represent room compatibility, room–time availability, and professor qualification (rather than temporal availability) constraints.
9. Finally, all numerical values in the small dataset were approximately doubled to create medium-sized datasets. To create large datasets, all numerical values in the medium dataset were approximately doubled.

It is important to emphasize that the structural assumptions adopted in the computational experiments—such as one-hour timeslots and a fixed number of daily periods—reflect the configuration used for synthetic data generation and do not constitute inherent restrictions of the proposed model. The formulation is fully parameterized with respect to the sets of days, timeslots, courses,

and institutional constraints. Consequently, the model can be readily adapted to alternative academic structures, including block scheduling systems, varying lecture durations (e.g., 50-min or 90-min sessions), or institutions with different daily period allocations. This flexibility ensures that the proposed framework is generalizable across diverse higher education settings.

The use of synthetically generated datasets is motivated by the absence of real-world timetabling data that jointly represent traditional courses and metaverse-based instructional activities within medical education. At the time of this study, metaverse courses have not yet been formally embedded into institutional scheduling systems in Turkey, making empirical datasets unavailable. Therefore, the synthetic data generation process was designed to mirror realistic institutional characteristics rather than to create arbitrary test cases. The adopted probability distributions reflect common structural properties observed in university settings, such as balanced course enrollments, limited availability of specialized rooms, and partial eligibility of professors and rooms. This controlled yet realistic design enables systematic evaluation of the proposed optimization framework across multiple problem scales while preserving practical relevance.

#### SA parameter setting

The efficiency and effectiveness of the SA algorithm are significantly influenced by the selection of its parameter values. The Taguchi DoE method was used to determine the optimal configuration for these parameters. This method allows a systematic analysis of the impact of each parameter and their interactions on performance metrics.

In this study, four key parameters of the SA algorithm were considered: the initial temperature coefficient, the cooling rate, the maximum number of iterations, and the number of iterations per temperature level. For each parameter, two levels were defined as follows: initial temperature coefficient $(k = 1, 10)$, cooling rate $\alpha = 0.99, 0.999)$, maximum number of iterations $(t_{max} = 1000, 2000)$, and maximum number of inner loop iterations $(i_{max} = 10, 20)$. The levels of the two-level factorial design are shown in Table 6. The parameter space is explored using the L16 orthogonal array, which allows both main and interaction effects to be investigated efficiently. The experiments were performed on a large number of problem instances with different scales (e.g., $n = 27, k = 23; n = 27, k = 25; n = 55, k = 55;$ and $n = 111, k = 103$). Each configuration was evaluated in terms of the average relative error (AVE) as the primary response variable. This approach aims to identify the most effective parameter combinations.

The results of the Taguchi design analysis revealed an excellent model fit, with an $R^2$ value of 0.998, indicating that the selected parameters explained almost all of the variability in the AVE. A significant interaction between the cooling rate and the maximum number of iterations was observed ($p = 0.025$), suggesting that the effect of cooling rate on the solution quality was dependent on the iteration level. Other interactions, including those involving the initial temperature coefficient and the maximum number of inner loop iterations, were found to be statistically insignificant.

Following the analysis, the configuration producing the most beneficial performance in terms of both solution quality and computational time was found to be: an initial temperature coefficient of $k = 1$, a cooling rate of $\alpha = 0.99$, a maximum number of iterations set to $t_{max} = 1000$, and a maximum number of inner loop iterations set to $i_{max} = 10$. These settings were adopted for all subsequent computational experiments using the SA algorithm in this study.

#### Experimental results

The performance of the GRAPE and SA algorithms was evaluated in terms of the average relative error and the computational time of the algorithms. The relative error was calculated for each test problem using Eq. (27).

$$\begin{aligned} RelatiiveError_{GRAPE} &= \frac{Z^* - Z_{GRAPE}}{Z^*}, \\ RelatiiveError_{SA} &= \frac{Z^* - Z_{SA}}{Z^*} \end{aligned} \tag{27}$$

where $Z^*$ is the optimal objective function value, $Z_{GRAPE}$ is the objective function value obtained by the proposed greedy algorithm, and $Z_{SA}$ is the objective function value produced by the Simulated Annealing algorithm for a test problem.

The optimal solutions are obtained using the binary integer programming formulation. The average relative errors and average computational times for GRAPE and SA are calculated by averaging 5 instances with different random seeds for each data set. The results in Table 7 show that GRAPE outperforms SA in terms of average computational times. However, SA method outperforms

**Table 6** Factor levels' coded values

| Level | Coded | Parameter | | | |
|---|---|---|---|---|---|
| | | $k$ | $\alpha$ | $t_{max}$ | $i_{max}$ |
| Low | −1 | 1 | 0.99 | 1000 | 10 |
| High | +1 | 10 | 0.999 | 2000 | 20 |

**Table 7** Results of the solution methods

| *n* | *m* | *k* | Average relative error | | # of optimal solutions | | Average computational time (s) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | GRAPE | SA | GRAPE | SA | Optimal | GRAPE | SA |
| 20 | 7 | 17 | 0.000 | 0.000 | 5 | 5 | 15.832 | 0.001 | 1.617 |
| | | 22 | 0.070 | 0.003 | 2 | 4 | 23.935 | 0.001 | 1.208 |
| | | 27 | 0.287 | 0.000 | 0 | 5 | 48.316 | 0.001 | 0.860 |
| 41 | 14 | 46 | 0.030 | 0.010 | 2 | 2 | 273.726 | 0.002 | 2.397 |
| | | 50 | 0.186 | 0.045 | 0 | 0 | 295.417 | 0.003 | 2.400 |
| | | 55 | 0.335 | 0.009 | 0 | 0 | 464.415 | 0.016 | 2.189 |
| 83 | 28 | 93 | 0.072 | 0.028 | 1 | 1 | 2769.207 | 0.024 | 5.896 |
| | | 101 | 0.222 | 0.028 | 0 | 0 | 2570.479 | 0.022 | 5.379 |
| | | 111 | 0.457 | 0.021 | 0 | 0 | 4513.282 | 0.035 | 4.954 |

**Table 8** Sensitivity analysis under ± 10% perturbation of perceived usefulness weight

| *n* | *m* | *k* | −10% Perturbation of perceived usefulness | | | | + 10% Perturbation of perceived usefulness | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Average objective deviation (%) | | Average assignment change (%) | | Average objective deviation (%) | | Average assignment change (%) | |
| | | | GRAPE | SA | GRAPE | SA | GRAPE | SA | GRAPE | SA |
| 20 | 7 | 17 | 0.21 | 0.21 | 0.00 | 0.00 | 0.24 | 0.24 | 0.00 | 0.00 |
| | | 22 | 0.05 | 0.44 | 0.00 | 14.29 | 0.11 | 0.49 | 0.00 | 8.57 |
| | | 27 | 0.07 | 0.07 | 0.00 | 54.29 | 0.17 | 1.33 | 0.00 | 68.57 |
| 41 | 14 | 46 | 0.05 | 0.66 | 0.00 | 10.00 | 0.10 | 0.85 | 2.86 | 11.43 |
| | | 50 | 0.09 | 1.78 | 0.00 | 51.43 | 0.13 | 2.91 | 0.00 | 45.71 |
| | | 55 | 0.04 | 4.10 | 0.00 | 77.14 | 0.08 | 2.04 | 0.00 | 75.71 |
| 83 | 28 | 93 | 0.09 | 1.46 | 0.71 | 23.57 | 0.15 | 0.87 | 0.00 | 24.29 |
| | | 101 | 0.12 | 1.14 | 0.00 | 57.14 | 0.15 | 1.26 | 6.43 | 60.71 |
| | | 111 | 0.02 | 1.26 | 5.71 | 90.71 | 0.08 | 1.14 | 0.00 | 87.14 |

GRAPE in terms of both lower average relative error values and frequency of reaching the optimal solution. This becomes more evident especially in datasets where the problem size increases (e.g. $n = 83$, $m = 28$, $k = 111$). Another remarkable finding is that both heuristics provide significant advantages in terms of computational time compared to optimal solutions. While it may take hours to obtain optimal solutions in large data sets, solutions can be produced in seconds with GRAPE and SA. It is also observed that GRAPE produces high relative error values in some cases (e.g. $AverageRelativeError_{GRAPE} = 0.457$), indicating that GRAPE tends to deviate from the optimality at larger problem sizes, but may still be suitable for obtaining a fast initial solution due to its low computational cost. In general, the SA algorithm can be considered as a more reliable approach in terms of solution quality and closeness to the optimum solution, while the GRAPE algorithm has the potential to produce low-cost initial solutions.

#### Sensitivity analysis of behavioral weights

To evaluate the robustness of the proposed SEM–AHP–optimization framework, a sensitivity analysis was conducted by perturbing the most influential behavioral construct—perceived usefulness—by ± 10%. This construct was selected due to its dominant weight in the AHP aggregation.

For each perturbation scenario, the optimization procedures (GRAPE and SA) were re-executed across all problem scales. The analysis assessed two indicators:

(i) percentage deviation in the objective function value relative to the baseline solution, and
(ii) percentage of metaverse course assignments that changed.

Table 8 summarizes the results.

Across all problem sizes, objective function deviations remained small. For GRAPE, average deviations were generally below 0.2%, while SA deviations remained below approximately 4% even in the largest instances. These findings indicate strong numerical stability of the scheduling objective with respect to moderate variations in behavioral weights.

Assignment-level changes were more pronounced in larger instances. In particular, when the number of professors increased, a higher proportion of course assignments shifted between solutions. However, these structural changes occurred without substantial deterioration in objective value.

This behavior suggests the presence of multiple near-optimal allocation structures. As the number of professors increases, normalized readiness weights become more closely distributed, leading to alternative schedules with comparable total weighted scores. Consequently, small weight perturbations may alter assignment configurations while preserving overall solution quality.

Overall, the results demonstrate that the proposed framework is robust at the objective-performance level, while exhibiting structural flexibility consistent with large-scale combinatorial scheduling environments.

## Discussion

The solution of the UCTTP is becoming increasingly difficult due to factors such as the specific needs of each educational institution and the increasing number of courses, lecturers and classrooms. Models developed for solving these problems sometimes take a very long time and sometimes become impossible to solve in large-scale problems where a large number of variables and constraints need to be considered together. Therefore, alternative solution methods such as heuristic and metaheuristic algorithms are employed to reduce computation time and obtain feasible results. Moreover, the integration of emerging technologies such as the metaverse into course content further complicates the timetabling problem not only from a computational perspective but also in terms of pedagogical and technological alignment. In this study, both of these issues are taken into account by specifically addressing the problem of course scheduling in the context of medical education, where the adoption of metaverse technologies is expected to become increasingly widespread. To this end, a more comprehensive and context-sensitive scheduling framework is created by integrating the behavioral intentions of medical educators towards the use of metaverse into the model.

The computational study conducted in this research is designed to evaluate the effectiveness of the proposed GRAPE and SA in solving the integrated UCTTP that includes both regular and metaverse courses. The performance of each algorithm is evaluated with synthetic datasets of different sizes and the results are evaluated in terms of average relative error, number of optimal solutions reached and computation time. The results reveal that SA achieves lower relative error values compared to GRAPE, especially for medium and large-scale samples. For example, on a dataset with $n = 83$, $m = 28$ and $k = 111$, SA achieved an AVE of 0.021 compared to GRAPE's significantly higher error of 0.457. Although GRAPE exhibits higher relative deviation in the largest problem instance, this outcome reflects the increased combinatorial complexity of large-scale timetabling scenarios rather than infeasibility. In this instance, the number of professors is nearly equal to the number of courses, resulting in a highly constrained allocation environment. GRAPE is designed to prioritize rapid generation of a feasible timetable satisfying all hard constraints, and therefore may sacrifice optimality in large instances. The subsequent SA phase significantly reduces this deviation by refining the initial solution. The tested instance sizes were calibrated to reflect the typical scale of medical schools in Turkey; however, substantially larger institutional settings would require additional computational calibration and parameter tuning, as is common in NP-hard scheduling environments. This performance trend confirms the robustness of SA in navigating highly constrained solution spaces, consistent with prior research demonstrating its effectiveness in complex timetabling environments (e.g., [17]). This finding is also consistent with the results of the bibliometric analysis, which identified SA as one of the most frequently applied metaheuristic techniques in the literature. In contrast, GRAPE has produced almost instantaneous solutions, even for large problems, with computation times often below 0.05 s compared to SA's runtime of a few seconds. While this supports the use of the GRAPE for generating fast initial solutions, it seems that the tendency to deviate from optimality is greater as the problem size increases. However, the use of the GRAPE as a suitable initial solution generator for metaheuristic approaches is also widely supported in the literature (e.g., [21]). These computational findings are particularly relevant in medical education contexts, where curriculum planning must balance technological innovation with institutional constraints and faculty readiness.

Taguchi DoE was used to determine the optimal combination of SA algorithm parameters. The model exhibited a very high coefficient of determination ($R2 = 0.998$), indicating that the selected parameters explain almost all the variance in AVE. This high explanatory power should be interpreted in light of the controlled experimental design: the response variable (AVE) is deterministically generated by the SA algorithm under predefined parameter settings rather than derived from noisy empirical observations. Therefore, the high R2 does not indicate overfitting in the traditional statistical sense, but rather reflects the structured nature of the orthogonal Taguchi design and the dominant influence of specific parameters on solution quality. A statistically significant interaction effect was found between the cooling rate and the maximum number of iterations, indicating that these parameters jointly influence the solution quality. This finding supports previous research on adaptive metaheuristics where parameter tuning is critical to optimize algorithmic performance (e.g., [26, 27]). Moreover, the selected parameter configuration was validated across multiple problem scales, indicating stable parameter sensitivity across the tested instance scales. Nevertheless, for

substantially larger or structurally different timetabling environments, re-tuning or adaptive parameter control strategies may be required. These results highlight the importance of integrating experimental design techniques into optimization processes while acknowledging the contextual scope of parameter calibration.

The sensitivity analysis further confirms that the proposed framework is not excessively dependent on exact behavioral weight values. While assignment configurations may vary under moderate weight perturbations—particularly in large-scale instances—the objective deviations remain minimal. This indicates that the model maintains performance stability despite the presence of multiple near-optimal solutions, reflecting realistic flexibility in faculty allocation decisions. The current sensitivity analysis focuses on ± 10% perturbations of the single most influential construct (perceived usefulness). Future work could extend this analysis to multi-parameter perturbation scenarios, where multiple behavioral constructs are varied simultaneously, as well as to more extreme perturbation magnitudes (e.g., ± 25% or ± 50%). Such analyses would provide a more comprehensive assessment of the robustness boundaries of the proposed framework and further support the generalizability of the findings.

An important contribution of the computational analysis is the demonstration of the feasibility and effectiveness of integrating the professor weights obtained from the SEM through AHP. This integration results in more realistic scheduling outcomes by prioritizing professors with a higher intention to teach in metaverse environments. The objective function that maximizes assignments based on weights represents a methodological innovation not found in previous scheduling studies by directly linking the empirical findings of behavioral intention modeling with optimization results.

These findings support the main goal of the study by confirming that integrating behavioral intention modeling into the scheduling process yields more informed and effective course allocation decisions, especially for emerging metaverse-based applications in medical education.

#### Theoretical implications

This work advances the theoretical landscape by combining combinatorial optimization with human-centered behavioral factors. The integration of SEM-derived constructs as criteria in an AHP framework and their subsequent inclusion as parameters in a binary integer programming model represents a significant methodological extension. In this way, the study not only addresses operational efficiency but also takes into account psychological variables that measure individuals' intentions to use technology. The combined consideration of this psychological and operational approach will be noteworthy given institutional trends towards immersive learning technologies that are increasingly used in medical education. Previous programming literature has largely ignored the inclusion of such latent psychological constructs in optimization problems. This work is thus intended to provide a foundation for future models that aim to incorporate faculty or student behavioral characteristics into scheduling or resource allocation mechanisms.

#### Practical implications

The proposed model provides significant practical benefits for academic institutions seeking to integrate technology-based courses such as metaverse applications into their curricula. By incorporating behavioral intention weights into the scheduling process, the model enables metaverse courses to be assigned to faculty members who are pedagogically and psychologically prepared to teach them. This facilitates smoother adoption of immersive technologies and reduces potential resistance during implementation. Furthermore, the hybrid scheduling framework accommodates both traditional academic constraints and emerging technological requirements, including metaverse-compatible classrooms, simulation laboratories, and virtual or augmented reality infrastructure. As such, it offers a structured and scalable decision-support tool for institutions operating under limited technological and human resources.

From an operational perspective, the computational results provide important insights for administrators. Although GRAPE may exhibit higher relative deviation in large-scale instances, it consistently generates fully feasible timetables that satisfy all hard constraints within negligible computation time. In practical settings, this allows institutions to rapidly obtain an operational schedule suitable for administrative review and implementation. When higher solution quality is required—particularly in complex or highly constrained scenarios—the SA phase can be employed to significantly reduce deviation from optimality. Therefore, the proposed GRAPE–SA hybrid approach provides flexibility: institutions may prioritize rapid feasibility in time-critical contexts or enhanced optimality when strategic alignment and resource efficiency are primary concerns.

#### Limitations and future work

The study has some limitations. First, the current formulation adopts a single-objective structure focused on maximizing the assignment of metaverse courses to professors with higher behavioral intention scores. Although this objective directly operationalizes the behavioral modeling component of the study, additional institutional priorities—such as workload fairness, room utilization

efficiency, or balanced teaching distribution—are not simultaneously optimized. Second, the computational experiments rely on synthetically generated datasets. While the use of synthetic data enables controlled experimentation, scalability analysis, and systematic evaluation of algorithmic performance, real-world institutional timetabling data may involve additional administrative rules, policy-driven constraints, or context-specific practices that are not fully captured in the current model. The reliance on synthetic data is primarily due to the early adoption stage of metaverse-based courses in medical education, where such courses are not yet formally integrated into institutional scheduling systems, limiting the availability of real-world datasets. Third, although the model prioritizes course allocations based on professor-specific weights, the current human-centered formulation focuses exclusively on instructor-level behavioral readiness. Student preferences, student technological access, and cohort-level preparedness for metaverse-based learning environments were not incorporated into the optimization framework. While faculty readiness was considered the initial implementation driver in this study, effective metaverse adoption also requires student-level alignment, which remains an important area for further investigation. Fourth, the behavioral weights were derived from a SEM study conducted among medical professionals in the Turkish higher education context. Cultural, institutional, and disciplinary differences may influence the relative importance of behavioral constructs such as perceived usefulness or imagination. Therefore, the empirical weights used in this study should be interpreted as context-specific parameters rather than universally generalizable values. In addition, the AHP weighting procedure was implemented under a single decision-maker assumption to reflect the administrative structure observed in the empirical context. However, in many higher education institutions, timetabling decisions are made collaboratively by committees or multiple stakeholders. While the proposed framework is flexible and can accommodate group-based AHP formulations, this extension was not explicitly implemented in the present study. Finally, while using the Taguchi design provides efficient parameter tuning, other experimental design approaches could be explored for better generalization or a comparative analysis could be conducted.

Future work could extend this study in several directions. First, the model can be expanded to multi-objective formulations that simultaneously consider behavioral readiness and additional institutional criteria, allowing decision-makers to explore trade-offs among competing priorities. Second, student-level behavioral intention, technological access, and cohort readiness could be integrated into the scheduling model to develop a more comprehensive multi-stakeholder decision framework. Third, the SEM–AHP component may be replicated across different countries, disciplines, and institutional contexts to generate locally calibrated readiness weights, enabling cross-cultural validation and comparative analysis. Fourth, the behavioral weights derived from the SEM model could be dynamically updated over time to reflect faculty development initiatives or evolving technological proficiency, enabling more adaptive and responsive scheduling systems. The AHP component may also be extended to group decision-making settings by aggregating pairwise comparisons from multiple stakeholders. Fifth, future studies may incorporate additional training components and heterogeneous course structures, such as clinical rotations, simulation-based modules, laboratory sessions, and interdisciplinary formats. Integrating these diverse instructional formats would introduce further scheduling dimensions, including variable session lengths, sequential dependencies, and shared resource constraints. Finally, as metaverse-based educational practices become operationally established, validating the proposed framework using real-world institutional data will be a critical step in assessing its practical effectiveness and broader applicability.

## Conclusions

The field of medical education has advanced significantly with the integration of emerging technologies. While educational strategies continue to evolve in response to global developments, one of the most significant innovations in recent years has been the emergence of the metaverse. This technology has started a new era in medical education by providing immersive and interactive learning experiences. These developments also necessitate the reorganization of course scheduling processes within medical faculties.

In this study, a binary integer linear programming model is developed for the UCTPP, in which metaverse and regular courses are considered together. Furthermore, various solution methods are proposed to solve the model. The proposed method consists of multiple stages. First, based on Damar and Koksalmis [15], the constructs from the SEM model are incorporated into the AHP model as criteria, and the resulting weights are integrated as parameters into the objective function of the optimization model. These weights represent professors' intentions to use metaverse technologies, and the model aims to prioritize the assignment of professors with high intention levels to metaverse courses. Subsequently, a GRAPE is developed to solve the scheduling problem, and this algorithm is used as the initial solution for the SA method. To enhance the efficiency of the SA algorithm, optimal parameter values are determined using the Taguchi DoE.

The current model assumes that pairwise comparisons in the AHP stage are conducted by a single decision maker. However, in institutional settings, these evaluations may involve multiple decision makers. Future implementations could incorporate group decision-making mechanisms that support collaborative input from faculty committees or administrative managers. Additionally, the model could be extended to include fuzzy AHP, which allows for the representation of uncertainty and ambiguity in preference judgments. Such enhancements would improve the realism, flexibility, and institutional applicability of the decision-making process within the model.

The solution of UCTTP becomes increasingly complex as the problem size and the variety of institutional requirements grow. This is supported by the experimental findings obtained in this study. An analysis of the solution table reveals that while optimal solutions can be achieved within the defined constraints, solution times increase significantly as the problem size increases. The experimental results also highlight the importance of the proposed heuristic algorithms in addressing this complexity. While the GRAPE is highly effective in generating rapid initial solutions, the SA algorithm significantly enhances solution quality and approaches optimality, particularly in larger and more constrained instances. Together, these two methods offer both computational efficiency and high-quality outcomes, making them a practical and scalable solution for real-world academic scheduling problems. However, the results also indicate that GRAPE alone may exhibit higher relative deviations in very large and tightly constrained problem instances. In such cases, the incorporation of the Simulated Annealing refinement phase becomes essential to achieve near-optimal solution quality. Accordingly, the GRAPE–SA hybrid structure provides a more robust and effective solution approach by combining rapid feasibility generation with solution refinement. As immersive technologies such as the metaverse become increasingly integrated into medical education, the ability to design adaptive, human-centered, and institutionally feasible programs will become even more critical. The proposed framework presents a proactive and practical roadmap for meeting these emerging needs.

**Abbreviations**

| | |
|---|---|
| UCTTP | University Course Timetabling Problem |
| VR | Virtual Reality |
| AR | Augmented Reality |
| SE | Simulation-based Education |
| AHP | Analytic Hierarchy Process |
| SEM | Structural Equation Modeling |
| SA | Simulated Annealing |
| GA | Genetic Algorithm |
| TS | Tabu Search |
| PSO | Particle Swarm Optimization |
| HS | Harmony Search |
| GRAPE | Greedy Reassignment and Assignment for Professor Equity |
| DoE | Design of Experiments |
| PW | Professor Weight |
| AVE | Average Relative Error |

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**Authors' contributions**
SD generated the data, formed the model, performed the analyses, and drafted the manuscript. GHK supervised the study, contributed to the study design and model development, and provided guidance throughout the research process. Both authors interpreted the results, reviewed, and approved the final manuscript.

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**Data availability**
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

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Not applicable.

**Consent for publication**
The authors have read and approved the manuscript in its current version.

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